

Thermodynamics and Statistical Physics

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Homework 5

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Scaling argument for the ideal-gas density of states ***

Consider an ideal gas of N identical particles with Hamiltonian

$$H(\mathbf{p}_1, \dots, \mathbf{p}_N) = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m}.$$

(a) Show that the Hamiltonian satisfies the scaling relation

$$H(\lambda \mathbf{p}_1, \dots, \lambda \mathbf{p}_N) = \lambda^2 H(\mathbf{p}_1, \dots, \mathbf{p}_N).$$

(b) Let

$$\Gamma(E) = \int_{H < E} d^{3N}p.$$

Using the scaling property, show that

$$\Gamma(E) \propto E^{\frac{3N}{2}}.$$

(c) Deduce that the phase-space volume of the energy shell satisfies

$$\Omega(E, V, N, \Delta E) \propto V^N E^{\frac{3N}{2}-1} \Delta E,$$

for small ΔE .

(d) Using $S = \log \Omega$, show that

$$S(E, V, N) = N \log V + \frac{3N}{2} \log E + \text{const}$$

up to subextensive corrections. Explain why the factor ΔE does not affect thermodynamic relations.

Solution

Part (a)

The Hamiltonian for an ideal gas of N identical particles is given by:

$$H(\mathbf{p}_1, \dots, \mathbf{p}_N) = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m}.$$

If we scale each momentum vector by a constant factor λ , i.e., $\mathbf{p}_i \rightarrow \lambda \mathbf{p}_i$, the Hamiltonian becomes:

$$H(\lambda \mathbf{p}_1, \dots, \lambda \mathbf{p}_N) = \sum_{i=1}^N \frac{(\lambda \mathbf{p}_i)^2}{2m} = \lambda^2 \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m} = \lambda^2 H(\mathbf{p}_1, \dots, \mathbf{p}_N).$$

This proves the scaling relation.

Part (b)

The momentum volume of the phase space up to energy E is defined as:

$$\Gamma(E) = \int_{H(\mathbf{p}) < E} d^{3N}p.$$

Let us introduce a change of variables to scale out the energy E . We define a dimensionless momentum $\mathbf{q}_i$ such that $\mathbf{p}_i = \sqrt{E} \mathbf{q}_i$. From the scaling relation derived in part (a), the condition $H(\mathbf{p}_1, \dots, \mathbf{p}_N) < E$ becomes:

$$H(\sqrt{E} \mathbf{q}_1, \dots, \sqrt{E} \mathbf{q}_N) < E \implies E \cdot H(\mathbf{q}_1, \dots, \mathbf{q}_N) < E \implies H(\mathbf{q}_1, \dots, \mathbf{q}_N) < 1.$$

The Jacobian of this transformation from $\mathbf{p}$ to $\mathbf{q}$ for a $3N$ -dimensional space introduces a factor of $(\sqrt{E})^{3N} = E^{3N/2}$. Thus, the integration measure becomes $d^{3N}p = E^{3N/2} d^{3N}q$. Substituting this into the integral yields:

$$\Gamma(E) = \int_{H(\mathbf{q}) < 1} E^{\frac{3N}{2}} d^{3N}q = E^{\frac{3N}{2}} \int_{H(\mathbf{q}) < 1} d^{3N}q.$$

The remaining integral is simply the volume of a $3N$ -dimensional unit sphere (weighted by masses), which is a constant completely independent of E . Therefore, we conclude that:

$$\Gamma(E) \propto E^{\frac{3N}{2}}.$$

Part (c)

The total phase-space volume $\Sigma(E, V, N)$ enclosed by the energy surface E includes the spatial integrals. For an ideal gas, the particles do not interact, so integrating over the spatial coordinates yields the volume V for each particle, giving a factor of V^N . Thus:

$$\Sigma(E, V, N) \propto V^N \Gamma(E) \propto V^N E^{\frac{3N}{2}}.$$

The phase-space volume of the energy shell $\Omega(E, V, N, \Delta E)$ of small width ΔE is the derivative of the total enclosed volume with respect to energy, multiplied by the thickness ΔE :

$$\Omega(E, V, N, \Delta E) = \frac{\partial \Sigma}{\partial E} \Delta E \propto \frac{\partial}{\partial E} \left(V^N E^{\frac{3N}{2}} \right) \Delta E \propto V^N \left(\frac{3N}{2} \right) E^{\frac{3N}{2}-1} \Delta E.$$

Absorbing the constant $3N/2$ into the proportionality, we get:

$$\Omega(E, V, N, \Delta E) \propto V^N E^{\frac{3N}{2}-1} \Delta E.$$

Part (d)

The Boltzmann entropy is defined as $S = \log \Omega$. Using the result from part (c), we have:

$$\begin{aligned} S(E, V, N) &= \log \left(C \cdot V^N E^{\frac{3N}{2}-1} \Delta E \right) \\ &= N \log V + \left(\frac{3N}{2} - 1 \right) \log E + \log \Delta E + \text{const.} \end{aligned}$$

For a macroscopic system, the number of particles $N \sim 10^{23}$, meaning $3N/2 \gg 1$, allowing us to approximate $\frac{3N}{2} - 1 \approx \frac{3N}{2}$. The term $\log \Delta E$ is an arbitrary parameter chosen for the shell thickness. However, thermodynamic relations are determined by the derivatives of the entropy with respect to extensive state variables (E, V, N). For instance, the temperature is given by $\frac{1}{T} = \frac{\partial S}{\partial E}$. Because $\log \Delta E$ is a constant or at most grows logarithmically (e.g., if $\Delta E \propto \sqrt{N}$), its derivative with respect to macroscopic energies or volumes is vanishingly small. Therefore, the $\log \Delta E$ term constitutes a subextensive correction that vanishes in the thermodynamic limit, meaning it has no effect on the derived macroscopic thermodynamic relations. Thus, dropping the subextensive terms, we find:

$$S(E, V, N) = N \log V + \frac{3N}{2} \log E + \text{const.}$$

Dependence on the shell thickness ★★ ★

Let

$$\Omega(E, \Delta E) = \int_E^{E+\Delta E} \Omega(E') dE', \quad \Omega(E') = e^{S(E')}.$$

Assume that the entropy $S(E)$ is smooth and expand it around E .

(a) Show that

$$S(E') = S(E) + \beta(E' - E) - \frac{(E' - E)^2}{2T^2C} + \dots,$$

where

$$\beta = \frac{\partial S}{\partial E} = \frac{1}{T}, \quad \frac{\partial^2 S}{\partial E^2} = -\frac{1}{T^2C},$$

and C is the heat capacity.

(b) Keeping only the linear term in the expansion, show that

$$\Omega(E, \Delta E) \approx \Omega(E) \frac{e^{\beta\Delta E} - 1}{\beta}.$$

(c) Show that when $\beta\Delta E \ll 1$, this reduces to

$$\Omega(E, \Delta E) \approx \Omega(E)\Delta E.$$

(d) (challenging) Using the linear approximation in part (b), show that the weight of energy levels within the shell can be written as

$$\Omega(E') \propto e^{\beta(E'-E)}.$$

Interpret this result and explain why a microcanonical ensemble with subextensive shell thickness ΔE can approximate a canonical ensemble. Show that the corresponding shell entropy

$$S_{\text{shell}}(E, \Delta E) = \log \Omega(E, \Delta E)$$

differs from $S(E)$ only by a subextensive correction.

Solution

Part (a)

Assume the entropy $S(E)$ is a smooth function. We perform a Taylor expansion of $S(E')$ around the energy E :

$$S(E') = S(E) + \left. \frac{\partial S}{\partial E} \right|_E (E' - E) + \frac{1}{2} \left. \frac{\partial^2 S}{\partial E^2} \right|_E (E' - E)^2 + \dots$$

From the fundamental thermodynamic relation, the inverse temperature is defined as:

$$\beta = \frac{1}{T} = \frac{\partial S}{\partial E}.$$

The second derivative is obtained using the chain rule and the definition of heat capacity $C = \frac{\partial E}{\partial T}$:

$$\frac{\partial^2 S}{\partial E^2} = \frac{\partial}{\partial E} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial E} = -\frac{1}{T^2 C}.$$

Substituting these derivatives back into the Taylor expansion yields:

$$S(E') = S(E) + \beta(E' - E) - \frac{(E' - E)^2}{2T^2 C} + \dots$$

Part (b)

Keeping only the linear term, the entropy is $S(E') \approx S(E) + \beta(E' - E)$. The density of states is $\Omega(E') = e^{S(E')}$. Substituting our linear approximation:

$$\Omega(E') \approx \exp[S(E) + \beta(E' - E)] = \Omega(E)e^{\beta(E' - E)}.$$

The total number of states in the shell of thickness ΔE is the integral over E' :

$$\Omega(E, \Delta E) = \int_E^{E+\Delta E} \Omega(E') dE' \approx \int_E^{E+\Delta E} \Omega(E)e^{\beta(E' - E)} dE'.$$

Let $x = E' - E$, so $dx = dE'$. The limits change from 0 to ΔE :

$$\Omega(E, \Delta E) \approx \Omega(E) \int_0^{\Delta E} e^{\beta x} dx = \Omega(E) \left[\frac{e^{\beta x}}{\beta} \right]_0^{\Delta E} = \Omega(E) \frac{e^{\beta \Delta E} - 1}{\beta}.$$

Part (c)

If the shell is very thin such that $\beta \Delta E \ll 1$, we can Taylor expand the exponential term: $e^{\beta \Delta E} \approx 1 + \beta \Delta E$. Substituting this into the result from part (b) gives:

$$\Omega(E, \Delta E) \approx \Omega(E) \frac{(1 + \beta \Delta E) - 1}{\beta} = \Omega(E) \frac{\beta \Delta E}{\beta} = \Omega(E) \Delta E.$$

This justifies the standard assumption that for extremely narrow shells, the volume is simply the density of states evaluated at E multiplied by the shell thickness.

Part (d)

From part (b), the linear approximation states that the density of states within the shell grows as:

$$\Omega(E') \approx \Omega(E)e^{-\beta E}e^{\beta E'} \implies \Omega(E') \propto e^{\beta E'}.$$

Interpretation: This exponential weight $e^{\beta E'}$ is mathematically identical to the Boltzmann factor that characterizes the canonical ensemble. It implies that within a microcanonical energy shell, the number of available microstates grows exponentially with energy. Because macroscopic systems have extremely sharply peaked energy distributions in contact with a heat bath, a microcanonical ensemble with a small, subextensive shell thickness inherently contains this canonical scaling within its bounds. The shell is thick enough to enclose the statistically significant fluctuations, hence mimicking the canonical ensemble perfectly over the relevant interval.

The shell entropy is:

$$\begin{aligned} S_{\text{shell}}(E, \Delta E) &= \log \Omega(E, \Delta E) \approx \log \left(\Omega(E) \frac{e^{\beta \Delta E} - 1}{\beta} \right) \\ &= S(E) + \log \left(\frac{e^{\beta \Delta E} - 1}{\beta} \right). \end{aligned}$$

The second term, $\log \left(\frac{e^{\beta \Delta E} - 1}{\beta} \right)$, depends only on the arbitrary shell parameter ΔE and the intensive parameter β . It does not scale linearly with the macroscopic particle number N (unlike $S(E) \propto N$). Thus, it is an $\mathcal{O}(1)$ or $\mathcal{O}(\log N)$ addition, making it a purely subextensive correction. In the thermodynamic limit, $S_{\text{shell}}(E, \Delta E) \approx S(E)$.

Maxwell distribution from the microcanonical ensemble ★★ ★

Consider an ideal gas of N identical particles in the microcanonical ensemble with total energy E .

- (a) Show that the marginal probability density of the momentum $\mathbf{p}_1$ of a single particle satisfies

$$f(\mathbf{p}_1) \propto \Omega\left(E - \frac{\mathbf{p}_1^2}{2m}, V, N - 1\right),$$

where $\Omega(E, V, N)$ is the density of states.

- (b) Using $S = \log \Omega$, rewrite the result as

$$f(\mathbf{p}_1) \propto \exp\left[S\left(E - \frac{\mathbf{p}_1^2}{2m}, V, N - 1\right)\right].$$

- (c) Assuming $N \gg 1$, expand the entropy to leading order in $\mathbf{p}_1^2/(2m)$, and show that

$$f(\mathbf{p}_1) \propto \exp\left(-\beta \frac{\mathbf{p}_1^2}{2m}\right), \quad \beta = \frac{\partial S}{\partial E}.$$

Normalize the distribution and recover the Maxwell distribution.

Solution

Part (a)

In the microcanonical ensemble, the total energy E is fixed, and the system is uniformly distributed over the allowed phase space. To find the marginal probability density $f(\mathbf{p}_1)$ of a single particle's momentum, we must integrate out the spatial coordinates of all N particles and the momenta of the remaining $N - 1$ particles.

Integrating over the positions gives a constant factor of V^N . The remaining $N - 1$ particles must satisfy the energy constraint given that particle 1 has momentum $\mathbf{p}_1$. The energy left for the $N - 1$ particles is $E' = E - \frac{\mathbf{p}_1^2}{2m}$. The integration over the remaining momenta $d^3p_2 \dots d^3p_N$ restricted to this energy is, by definition, the density of states for an ideal gas of $N - 1$ particles occupying volume V with total energy E' . Thus, the marginal probability density is proportional to this phase-space volume:

$$f(\mathbf{p}_1) \propto \Omega\left(E - \frac{\mathbf{p}_1^2}{2m}, V, N - 1\right).$$

Part (b)

By definition of the Boltzmann entropy, $\Omega = e^S$. Therefore, we can immediately rewrite the proportionality from part (a) in terms of the entropy of the $N - 1$ particle system:

$$f(\mathbf{p}_1) \propto \exp\left[S\left(E - \frac{\mathbf{p}_1^2}{2m}, V, N - 1\right)\right].$$

Part (c)

For a macroscopic gas where $N \gg 1$, the kinetic energy of a single particle is minuscule compared to the total macroscopic energy: $\frac{\mathbf{p}_1^2}{2m} \ll E$. We can perform a Taylor expansion of the entropy S around the total energy E :

$$S\left(E - \frac{\mathbf{p}_1^2}{2m}, V, N - 1\right) \approx S(E, V, N - 1) - \frac{\partial S}{\partial E} \left(\frac{\mathbf{p}_1^2}{2m}\right) + \dots$$

By definition in statistical mechanics, $\frac{\partial S}{\partial E} = \beta = \frac{1}{k_B T}$ (taking $k_B = 1$ gives $\beta = 1/T$). Because $S(E, V, N - 1)$ does not depend on $\mathbf{p}_1$, the term $e^{S(E, V, N - 1)}$ is just a normalization constant. Taking the exponential of the expanded entropy gives:

$$f(\mathbf{p}_1) \propto \exp\left(-\beta \frac{\mathbf{p}_1^2}{2m}\right).$$

To find the exact distribution, we must normalize $f(\mathbf{p}_1)$ such that $\int f(\mathbf{p}_1) d^3 p_1 = 1$. Let $f(\mathbf{p}_1) = C \exp\left(-\beta \frac{\mathbf{p}_1^2}{2m}\right)$. The integral over all momentum space is a standard 3D Gaussian integral:

$$\int_{\mathbb{R}^3} C \exp\left(-\frac{\beta}{2m}(p_{1x}^2 + p_{1y}^2 + p_{1z}^2)\right) d^3 p_1 = C \left(\sqrt{\frac{2m\pi}{\beta}}\right)^3 = C \left(\frac{2\pi m}{\beta}\right)^{\frac{3}{2}} = 1.$$

Solving for C gives $C = \left(\frac{\beta}{2\pi m}\right)^{\frac{3}{2}}$. Thus, we recover the Maxwell distribution for the momentum of a single particle:

$$f(\mathbf{p}_1) = \left(\frac{\beta}{2\pi m}\right)^{\frac{3}{2}} \exp\left(-\beta \frac{\mathbf{p}_1^2}{2m}\right).$$

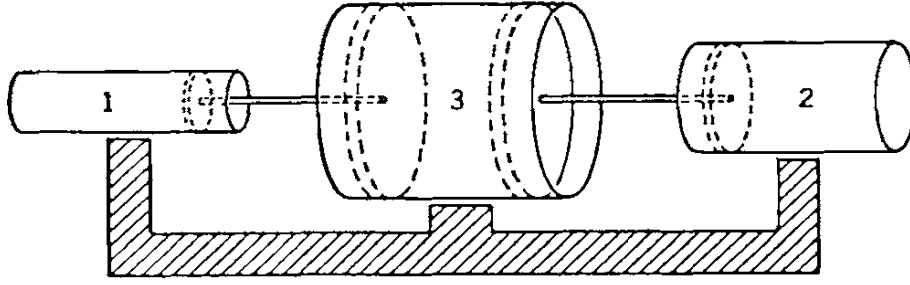


图 1: Three ideal gas subsystems (labeled 1, 2, and 3) confined by movable, frictionless pistons. Each gas occupies a volume V_i and is bounded by a piston with cross-sectional area A_i . The pistons are adiabatic, and the total length of the system is fixed: $\delta\ell_1 + \delta\ell_2 + \delta\ell_3 = 0$. The temperatures T_i of the subsystems may differ. Assume $A_1 : A_2 : A_3 = 1 : 2 : 3$.

Mechanical Equilibrium with Adiabatic Pistons and Unequal Temperatures
(adapted from Callen)*****

Consider three ideal gases (labeled 1, 2, and 3) confined in a one-dimensional arrangement by two movable, frictionless, and adiabatic pistons, as shown in Fig. 1. Each gas is enclosed in a rigid cylinder with cross-sectional area A_i , and the total length of the system is fixed:

$$\delta\ell_1 + \delta\ell_2 + \delta\ell_3 = 0.$$

The pistons are adiabatic and thermally insulating, so no heat or entropy can be exchanged between the gases. As a result, each gas may have a different temperature T_i . However, the system is allowed to reach **mechanical equilibrium**. To make the setup concrete, assume that the cross-sectional areas of the cylinders are in the ratio: $A_1 : A_2 : A_3 = 1 : 2 : 3$. Let $\delta\ell_i$ denote the infinitesimal displacement of the piston bounding gas i , and let the corresponding volume change be: $\delta V_i = A_i \delta\ell_i$.

- (A) Use the principle of virtual work to derive the condition for mechanical equilibrium among the three gases. What relation must the pressures P_i satisfy? *Hint: Mechanical equilibrium requires that the total virtual work vanishes.*
- (B) In the isothermal case (where the three gases are in thermal contact), mechanical equilibrium is determined by maximizing the total entropy. What condition replaces your result from part (A) in this case? How do the equilibrium conditions differ between the isothermal and adiabatic settings?
- (C) Explain why entropy maximization does not apply in this adiabatic setup. What is the key physical difference in the constraints that leads to different criteria for equilibrium? *Hint: In the adiabatic setup, can entropy be exchanged between the subsystems? Which macroscopic quantities are allowed to vary, and which are constrained?*

Solution

Part (A)

According to the principle of virtual work, a system is in mechanical equilibrium if the total virtual work done *by* the system is zero for any arbitrary infinitesimal displacement that respects the constraints. The total virtual work done by the three gases during infinitesimal volume changes δV_i is:

$$\delta W = P_1 \delta V_1 + P_2 \delta V_2 + P_3 \delta V_3.$$

The geometric constraint given is $\delta \ell_1 + \delta \ell_2 + \delta \ell_3 = 0$. Since $\delta V_i = A_i \delta \ell_i$, we can write $\delta \ell_i = \frac{\delta V_i}{A_i}$. Therefore, the constraint in terms of volumes is:

$$\frac{\delta V_1}{A_1} + \frac{\delta V_2}{A_2} + \frac{\delta V_3}{A_3} = 0.$$

To maximize the work subject to this constraint, we introduce a Lagrange multiplier λ :

$$\delta W - \lambda \left(\frac{\delta V_1}{A_1} + \frac{\delta V_2}{A_2} + \frac{\delta V_3}{A_3} \right) = 0.$$

Grouping terms for independent variations $\delta V_1, \delta V_2, \delta V_3$:

$$\left(P_1 - \frac{\lambda}{A_1} \right) \delta V_1 + \left(P_2 - \frac{\lambda}{A_2} \right) \delta V_2 + \left(P_3 - \frac{\lambda}{A_3} \right) \delta V_3 = 0.$$

For this to hold for arbitrary variations, the coefficients must individually vanish, yielding:

$$P_1 A_1 = P_2 A_2 = P_3 A_3 = \lambda.$$

This represents mechanical equilibrium: the forces ($F_i = P_i A_i$) exerted by the gases on the piston assembly must be equal to balance the system.

Part (B)

In the isothermal case (thermal contact), equilibrium is found by maximizing the total entropy $S_{\text{tot}} = S_1 + S_2 + S_3$. The fundamental thermodynamic identity is $dS_i = \frac{1}{T_i} dU_i + \frac{P_i}{T_i} dV_i$. We maximize S_{tot} subject to two constraints: 1) Energy conservation: $\delta U_1 + \delta U_2 + \delta U_3 = 0$. 2) Geometric length: $\frac{\delta V_1}{A_1} + \frac{\delta V_2}{A_2} + \frac{\delta V_3}{A_3} = 0$.

Using two Lagrange multipliers λ_U and λ_V :

$$\delta S_{\text{tot}} = \sum_{i=1}^3 \left(\frac{1}{T_i} \delta U_i + \frac{P_i}{T_i} \delta V_i \right) - \lambda_U \sum_{i=1}^3 \delta U_i - \lambda_V \sum_{i=1}^3 \frac{\delta V_i}{A_i} = 0.$$

Grouping variations in δU_i :

$$\left(\frac{1}{T_i} - \lambda_U \right) \delta U_i = 0 \implies T_1 = T_2 = T_3 = T.$$

Grouping variations in δV_i :

$$\left(\frac{P_i}{T_i} - \frac{\lambda_V}{A_i} \right) \delta V_i = 0 \implies \frac{P_i A_i}{T_i} = \lambda_V.$$

Since $T_1 = T_2 = T_3 = T$, we can multiply the volume condition by T to find:

$$P_1 A_1 = P_2 A_2 = P_3 A_3.$$

Differences: The pure mechanical equilibrium condition ($P_1 A_1 = P_2 A_2 = P_3 A_3$) is identical in both cases. However, in the isothermal case, entropy maximization introduces a strictly required thermal condition ($T_1 = T_2 = T_3$). In the adiabatic case from Part (A), the temperatures T_i are arbitrary and decouple from the mechanical balance.

Part (C)

Entropy maximization determines equilibrium by finding the configuration of internal, unconstrained parameters that results in the highest possible total entropy. In the strictly adiabatic setup, each gas is thermally isolated and undergoes reversible quasistatic expansions or compressions. This implies that the entropy of each subsystem is physically fixed: $\delta S_1 = \delta S_2 = \delta S_3 = 0$.

Because no entropy can be exchanged between subsystems and no new entropy is generated reversibly, the total entropy variation is identically zero ($\delta S_{\text{tot}} = 0$) for all allowed geometric displacements. There is no entropy gradient to drive the system toward equilibrium. Therefore, the principle of maximum entropy provides no information. Instead, the system's macroscopic variables S_i are strictly constrained, while the energies U_i and volumes V_i are allowed to vary but are deterministically coupled via $dU_i = -P_i dV_i$. The equilibrium criterion physically reverts to the minimization of total mechanical energy (or zero virtual work), governed entirely by Newton's laws for the coupled pistons.